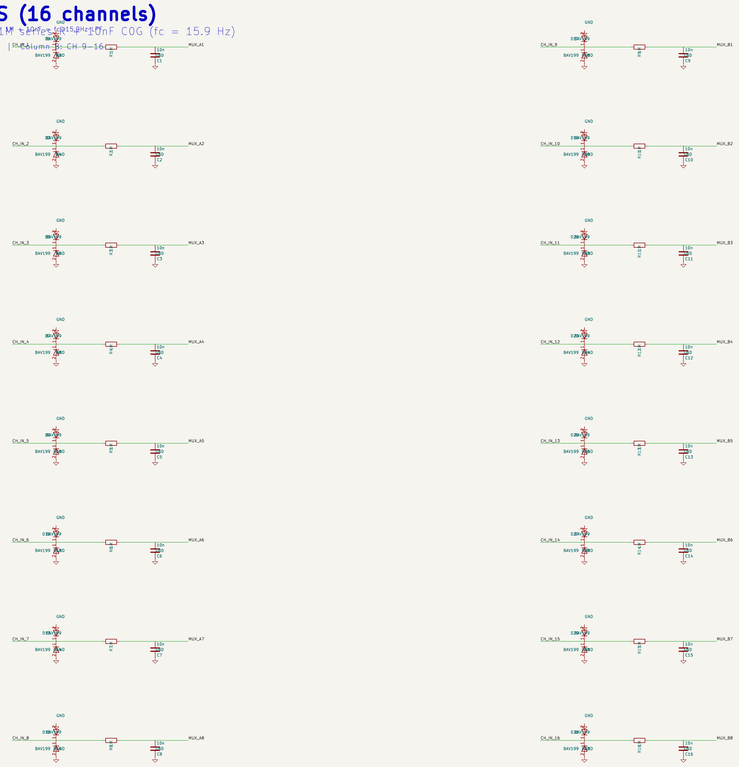
CircuitForge — 16-Channel Multiplexed Measurement System

ADuCM362 + ADA4530-1 TIA | 4 Relay-Selectable Ranges | 24-bit ADC

INPUT FILTERS (16 channels)

Per channel: BAV199 ESD clamp | 1M series | 10nF C0G (fc = 15.9 Hz)

Column A: CH 1-8 | Columns B: CH 9-16

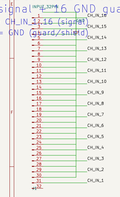


CONNECTOR

16 signal + 16 GND guard)

CH_IN_1-16 (signal)

GND (guard/shield)



signal, guard=driven, shield=GND

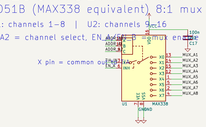
ANALOG MULTIPLEXER

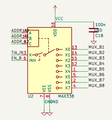
2x CD4051B (MAX338 equivalent), 8:1 mux

U1: channels 1-8 | U2: channels 9-16

ADDR_A0-A2 = channel select, EN_A/B = enable

X pin = common output





TRANSIMPEDANCE AMPLIFIER

U3: ADA4530-1 (20fA bias current)

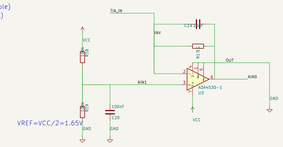
Rf/Cf: relay-selectable (10M / 100M / 1G+10pF / 10G+1pF)

VREF: 100k/100k divider = 1.65V mid-supply

Output: AIN0 (TIA) + AIN1 (VREF monitor)

Rf=1G default (relay-selectable)

Cf=10pF (stability, BW limit)



VREF=VCC/2=1.65V

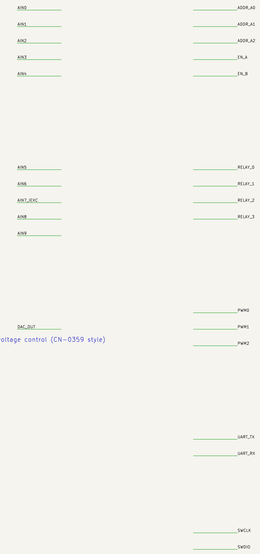
MCU — ADuCM362

ARM Cortex-M3, Dual 24-bit Sigma-Delta ADC

ADC0: AIN0-4 (TIA + AVDD monitor + guard) | ADC1: AIN5-9 (RTD 2/3/4-wire)

GPIO: Mux addr, Relay ctrl, PWM | UART + SWD debug

Internal 1.2V VREF (no external ref needed)

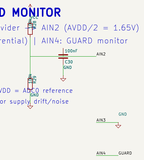


DAC_OUT: excitation voltage control (CN-0359 style)

AVDD MONITOR

R28/R29: 100k/100k divider | AIN2 (AVDD/2 = 1.65V)

AIN3: AGND ref (differential) | AIN4: GUARD monitor

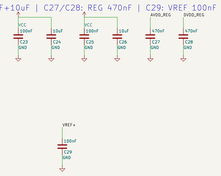


Critical: AVDD = ADC0 reference

Monitor for supply drift/noise

DECOUPLING CAPACITORS

C23/C24: AVDD 100nF+10uF | C25/C26: DVDD 100nF+10uF | C27/C28: REG 470nF | C29: VREF 100nF



RTD TEMPERATURE SENSOR

J2: 4-wire RTD connector (CN-0359 style auto-detect 2/3/4-wire)

AIN7: IEXC 600uA | AIN8/9: Kelvin sense | R30: 4.5k RREF

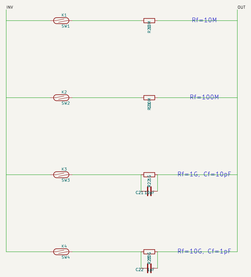
Temp logged with each channel reading for drift compensation

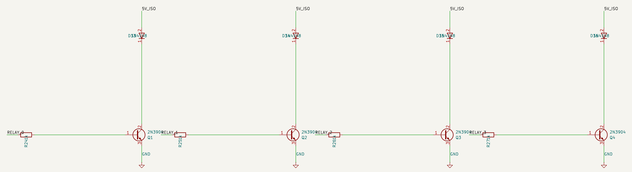
RELAY RANGE LADDER

4x 109P reed relay + 2N3904 NPN drivers

K1: 10M (120nA FS) | K2: 100M (12nA FS) | K3: 1G+10pF (1.2nA FS) | K4: 10G+1pF (120pA FS)

RELAY_0-3 GPIO from MCU | 5V_ISO coil supply | 1N4148 flyback protection





Rb=1k (Ic/hFE=5V/0.05A/100=0.5mA. 1k gives 3.3mA base drive)

D33-D36: 1N4148 flyback diodes (relay coil back-EMF protection)

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